**Quadruped Robot – Mechanical Design (First Task)**

**1. Degrees of Freedom**

- **8 DOF (2 DOF × 4 legs)**

**2. Body Shape & Structure**

The robot uses a **rectangular chassis** measuring approximately **120 mm (length) × 100 mm (width) × 40 mm (height)**, made from **3D-printed PLA** with 3 mm wall thickness. The rectangular shape was chosen because:

- It provides a **wide, symmetric support polygon** between the four legs, improving stability during the trot gait.
- Flat surfaces simplify mounting the electronics: the microcontroller and servo driver sit on the **top plate**, while the **2000 mAh Li-Po battery is mounted on the bottom plate** — keeping the heaviest component low to lower the center of gravity.
- The **two-plate design** (top + bottom plates connected by standoffs) makes assembly and maintenance easy and adds torsional rigidity.

The four hip servos are mounted at the **four corners of the chassis**, placing the legs as far apart as possible to maximize the support base.

**3. Leg Design (2 DOF per leg)**

Each leg has **two joints, each driven by one servo**:

| Joint | Function | Axis |
|---|---|---|
| **Hip (shoulder)** | Swings the leg forward/backward for stepping | Vertical rotation |
| **Knee (lift)** | Raises and lowers the foot off the ground | Horizontal rotation |

- Each leg consists of **two rigid PLA links**: an upper link (femur, **40 mm**) and a lower link (tibia, **40 mm**), giving the total moment arm of **L = 80 mm** used in the torque calculation below.
- The foot tip has a **rubber cap** to increase friction and prevent slipping on smooth floors.
- Legs are angled slightly outward (~10°) to widen the support polygon, directly addressing the tipping-over risk identified in Section 7.

**4. Servo Selection: MG90S (Metal Gear Micro Servo)**

**Chosen servo: MG90S** — one per joint, **8 servos total**.

**Justification:**

- **Torque:** The MG90S provides **2.2 kg·cm at 4.8 V** (2.5 kg·cm at 6 V). The calculated static requirement is **1.0 kg·cm per joint** (see Section 5), giving a **safety factor of ≈ 2.2×**, which covers dynamic loads (walking impacts, acceleration) not included in the static calculation.
- **Weight:** Only **13.4 g per servo** → 8 servos add ~107 g, keeping total mass within the 0.5 kg budget used in the calculation. A larger servo such as the MG996R (55 g each = 440 g for 8) would consume almost the entire mass budget.
- **Metal gears:** Unlike the cheaper SG90 (plastic gears), the MG90S withstands the repeated shock loads of walking without gear stripping.
- **Cost & availability:** Inexpensive (~$3–5) and widely available, suitable for a prototype.

**Comparison of candidate servos:**

| Option | Torque | Weight | Verdict |
|---|---|---|---|
| SG90 | 1.8 kg·cm | 9 g | ✗ Plastic gears fail under walking shocks; low safety margin |
| **MG90S** | **2.2 kg·cm** | **13.4 g** | ✓ **Best torque-to-weight, metal gears, ≈2× safety factor** |
| MG996R | 9–11 kg·cm | 55 g | ✗ Overkill; too heavy for a 0.5 kg robot |

**5. Preliminary Torque Calculation for One Joint**

**Given Data:**

- Total robot mass: $m = 0.5$ kg
- Gravitational acceleration: $g = 9.81$ m/s$^2$
- Leg length (moment arm): $L = 0.08$ m (80 mm)
- Number of legs supporting weight: 4

**Calculation Steps:**

$F_{total} = m \times g = 0.5 \times 9.81 = 4.905$ N

$F\_leg = F\_total / 4 = 4.905 / 4 = 1.226$ N

$\tau = F\_leg \times L = 1.226 \times 0.08 = 0.098$ N·m

$\tau = 0.098$ N·m × 10.2 = **1.0 kg·cm**

→ The selected MG90S (2.2 kg·cm) satisfies this requirement with a safety factor of ≈ 2.2.

**6. Stability & Center of Gravity (COG)**

**Location of Center of Gravity:**

- Horizontally: In the middle of the body (between the four legs).
- Vertically: As low as possible (near the legs) — achieved by mounting the battery on the bottom plate.

**7. Proposed Walking Gait**

| Phase | Moving Legs | Stationary Legs |
|---|---|---|
| 1 | Front-Left + Back-Right | Front-Right + Back-Left |
| 2 | Short pause for stability | All legs on ground |
| 3 | Front-Right + Back-Left | Front-Left + Back-Right |
| 4 | Repeat cycle | – |

**8. Expected Mechanical Problems**

| Problem | Cause | Proposed Solution |
|---|---|---|
| Tipping over | High center of gravity | Lower the body / widen support base |
| Short battery life | High servo power consumption | Use larger battery (2000 mAh Li-Po) |

---